

UNIVERSITY OF KELANIYA - SRI LANKA

Faculty of Science

BECS 21413 / BECS 21413 (R)

Analogue Electronics II

(Operational Amplifiers)

Detailed Model Solutions

July/August 2025 Examination Paper

These solutions answer **all seven questions**, even though the examination requires candidates to answer only five. Derivations are written in a marking-friendly sequence, and all circuit analyses assume an ideal op-amp unless a non-ideal effect is explicitly discussed.

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General assumptions and notation

For the standard ideal op-amp analysis used throughout this document:

- The open-loop gain is very large: $A_{OL} \rightarrow \infty$.
- The input resistance is very large, so $i_+ = i_- = 0$.
- With negative feedback and unsaturated operation, the differential input voltage is approximately zero, hence $V_+ \approx V_-$ (the “virtual short” condition).
- The output resistance is approximately zero.
- Supply rails and ideal DC voltage sources are AC ground when an AC equivalent circuit is formed.

Question 1 - Negative feedback and basic closed-loop amplifiers

1(a) Negative feedback concept

Definition and operation

Negative feedback means that a fraction of the output is returned to the input in a direction that *opposes* the applied signal. In a voltage-feedback op-amp circuit, the feedback signal is normally connected to the inverting input.

Let the op-amp open-loop gain be A and let a fraction β of the output be fed back. Then

$$V_f = \beta V_o, \quad V_d = V_s - V_f, \quad V_o = AV_d.$$

Therefore,

$$V_o = A(V_s - \beta V_o),$$

so

$$V_o(1 + A\beta) = AV_s.$$

The closed-loop gain is therefore

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}.$$

When $A\beta \gg 1$,

$$A_f \approx \frac{1}{\beta},$$

which shows that the gain is mainly fixed by the external feedback network rather than the uncertain open-loop gain of the op-amp.

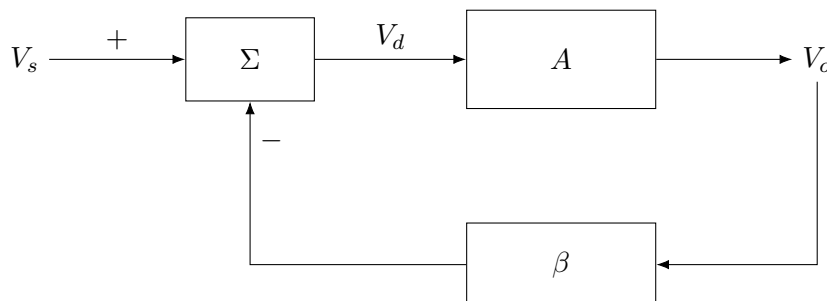


Figure 1: General negative-feedback system.

Important practical effects of negative feedback are:

- accurately controlled closed-loop gain;
- increased bandwidth;
- reduced distortion and sensitivity to component/device variations;
- high input resistance and low output resistance for many voltage-amplifier configurations;
- stable linear operation, provided the loop remains stable and the output does not saturate.

1(b) Inverting and non-inverting closed-loop amplifiers

Inverting amplifier

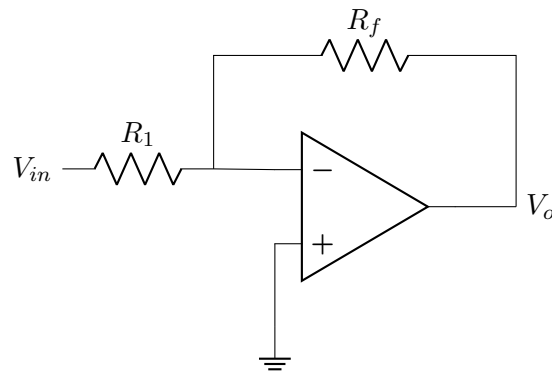


Figure 2: Basic inverting amplifier.

Since $V_+ = 0$, negative feedback gives $V_- \approx 0$. The inverting input is therefore a *virtual ground*. No current enters the op-amp input, so the current through R_1 must flow through R_f :

$$\frac{V_{in} - 0}{R_1} = \frac{0 - V_o}{R_f}.$$

Hence,

$$V_o = -\frac{R_f}{R_1} V_{in}$$

and the closed-loop voltage gain is

$$A_{v,inv} = \frac{V_o}{V_{in}} = -\frac{R_f}{R_1}.$$

The negative sign indicates a phase reversal of 180° .

Non-inverting amplifier

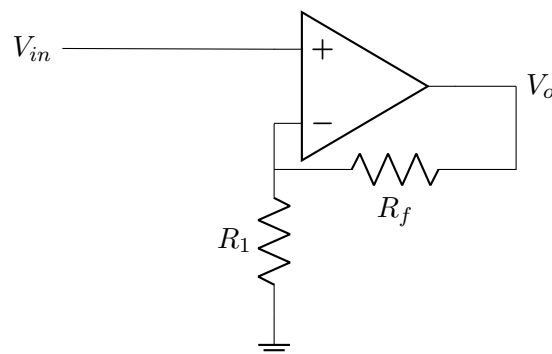


Figure 3: Basic non-inverting amplifier.

The input is applied to the non-inverting terminal, so

$$V_+ = V_{in}.$$

Negative feedback gives

$$V_- = V_+ = V_{in}.$$

The feedback divider produces

$$V_- = V_o \frac{R_1}{R_1 + R_f}.$$

Therefore,

$$V_{in} = V_o \frac{R_1}{R_1 + R_f},$$

so

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

and

$$A_{v,noninv} = 1 + \frac{R_f}{R_1}.$$

The gain is positive, so the output is in phase with the input.

Comparison

Configuration	Closed-loop gain	Input phase relation
Inverting	$-R_f/R_1$	180° phase reversal
Non-inverting	$1 + R_f/R_1$	In phase

1(c) Inverter and voltage follower

Unity-gain inverter

An inverter is obtained from the inverting amplifier by selecting

$$R_f = R_1.$$

Then

$$V_o = -V_{in}, \quad A_v = -1.$$

The output has the same magnitude as the input but opposite polarity.

Voltage follower

A voltage follower is the limiting case of the non-inverting amplifier. The output is connected directly to the inverting input, giving $\beta = 1$.

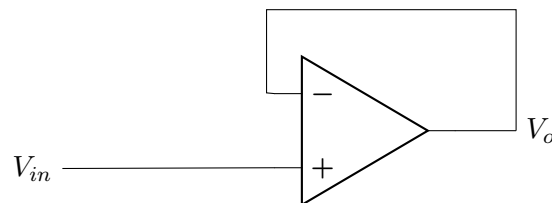


Figure 4: Voltage follower (unity-gain buffer).

Because $V_- = V_o$ and $V_+ = V_{in}$,

$$V_o = V_{in}, \quad A_v = 1.$$

The voltage follower is useful for impedance matching because it has a very high input resistance and a low output resistance.

1(d) Single-supply AC inverting amplifier

The circuit on page 1 of the paper uses two equal resistors to generate a mid-supply reference, input and output coupling capacitors, and an ordinary inverting feedback network.

1(d)(i) DC equivalent circuit and quiescent output level

For DC analysis:

- C_{in} and C_o are open circuits;
- the input source and the load are therefore disconnected from the amplifier;
- the equal divider resistors establish

$$V_+ = \frac{R}{R+R}V_B = \frac{V_B}{2}.$$

Since the input capacitor is open, no DC current flows through R_1 . Since op-amp input current is zero, no DC current flows through R_f either. Therefore there is no DC drop across R_f , and

$$V_o(DC) = V_- = V_+ = \frac{V_B}{2}.$$

Thus the internal op-amp output is biased at the centre of the available supply range:

$$V_{o,Q} = \frac{V_B}{2}.$$

The total instantaneous output before the coupling capacitor is

$$v_o(t) = \frac{V_B}{2} + v_{o,ac}(t).$$

For the inverting AC signal,

$$v_{o,ac}(t) = -\frac{R_f}{R_1}v_{in,ac}(t).$$

Hence the amplified AC waveform swings above and below the DC level $V_B/2$.

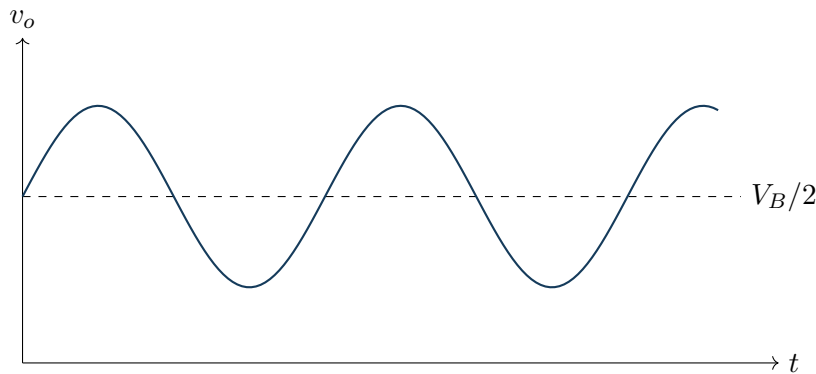


Figure 5: The AC output is centred on the quiescent level $V_B/2$.

1(d)(ii) AC equivalent circuit and load voltage

For the mid-band AC equivalent:

- C_{in} and C_o are treated as short circuits;
- the ideal DC supply V_B is AC ground;
- therefore both ends of the bias divider are at AC ground, so the non-inverting input is at AC ground;
- the amplifier becomes an ordinary inverting amplifier.

Therefore,

$$v_{o,ac} = -\frac{R_f}{R_1}v_{in,ac}.$$

The output coupling capacitor blocks the DC term $V_B/2$ but passes the AC term. Hence the load voltage is

$$v_L(t) = v_{o,ac}(t) = -\frac{R_f}{R_1}v_{in,ac}(t).$$

Thus only the AC signal appears across R_L ; the DC bias does not reach the load.

Exam-writing point

In the exam, clearly write “capacitors open for DC” and “capacitors short, ideal supplies at AC ground for mid-band AC”. These two statements justify the equivalent circuits and usually earn method marks.

Question 2 - Ideal and practical op-amp integrators

2(a) Show that the true integrator integrates the input

For the ideal inverting integrator, $V_+ = 0$, so $V_- \approx 0$. The current through the input resistor is

$$i_R = \frac{V_{in}}{R_1}.$$

No current enters the op-amp input. Therefore the same current flows through the feedback capacitor. Taking current from the summing node toward the output,

$$i_C = C_f \frac{d(V_- - V_o)}{dt} = C_f \frac{d(0 - V_o)}{dt} = -C_f \frac{dV_o}{dt}.$$

Equating currents,

$$\frac{V_{in}}{R_1} = -C_f \frac{dV_o}{dt}.$$

Therefore,

$$\frac{dV_o}{dt} = -\frac{1}{R_1 C_f} V_{in}(t).$$

Integrating from t_0 to t ,

$$V_o(t) - V_o(t_0) = -\frac{1}{R_1 C_f} \int_{t_0}^t V_{in}(\tau) d\tau.$$

Hence,

$$V_o(t) = V_o(t_0) - \frac{1}{R_1 C_f} \int_{t_0}^t V_{in}(\tau) d\tau.$$

If the initial capacitor voltage is zero,

$$V_o(t) = -\frac{1}{R_1 C_f} \int_0^t V_{in}(\tau) d\tau.$$

The output is therefore the negative time integral of the input.

2(b) Response to a constant 1 V input

Let

$$V_{in}(t) = 1 \text{ V}, \quad t \geq 0.$$

Then

$$V_o(t) = V_o(0) - \frac{1}{R_1 C_f} \int_0^t 1 d\tau = V_o(0) - \frac{t}{R_1 C_f}.$$

For $V_o(0) = 0$,

$$V_o(t) = -\frac{t}{R_1 C_f}.$$

This is a negative-going linear ramp. Its gradient is

$$\frac{dV_o}{dt} = -\frac{1}{R_1 C_f} \text{ V s}^{-1}.$$

More generally, for a constant input V_{DC} ,

$$\frac{dV_o}{dt} = -\frac{V_{DC}}{R_1 C_f}.$$

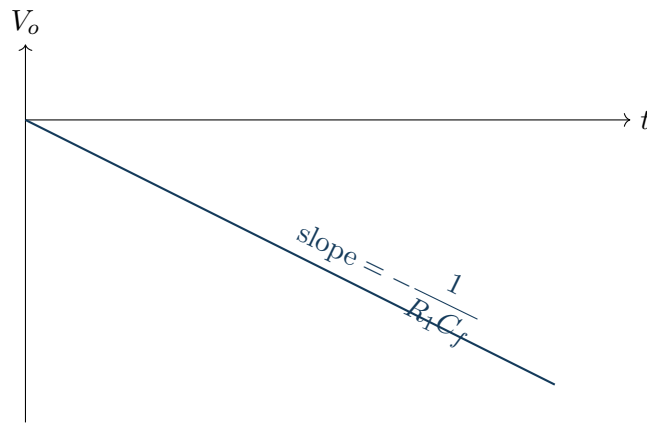


Figure 6: Output of the true integrator for a positive constant input.

2(c) Transfer function and low-frequency design problem

Using impedances,

$$Z_{in} = R_1, \quad Z_f = \frac{1}{sC_f}.$$

For an inverting op-amp,

$$H(s) = \frac{V_o(s)}{V_{in}(s)} = -\frac{Z_f}{Z_{in}}.$$

Therefore,

$$H(s) = -\frac{1}{sR_1C_f}.$$

At $s = j\omega$,

$$H(j\omega) = -\frac{1}{j\omega R_1C_f} = \frac{j}{\omega R_1C_f}.$$

Thus

$$|H(j\omega)| = \frac{1}{\omega R_1C_f}, \quad \angle H = +90^\circ.$$

The magnitude increases without limit as $\omega \rightarrow 0$. In particular, the ideal integrator has infinite gain at DC. Therefore very small DC errors such as input offset voltage, input bias-current effects, and a small DC component in the signal are integrated continuously. The output then drifts toward one supply rail and saturates.

Main design issue

$$\omega \rightarrow 0 \Rightarrow |H| \rightarrow \infty$$

Consequently, a true integrator is normally unsuitable as a practical stand-alone circuit unless a reset path or finite DC-feedback path is provided.

2(d) Practical integrator and its transfer function

A practical integrator is obtained by connecting a resistor R_f in parallel with the feedback capacitor C_f .

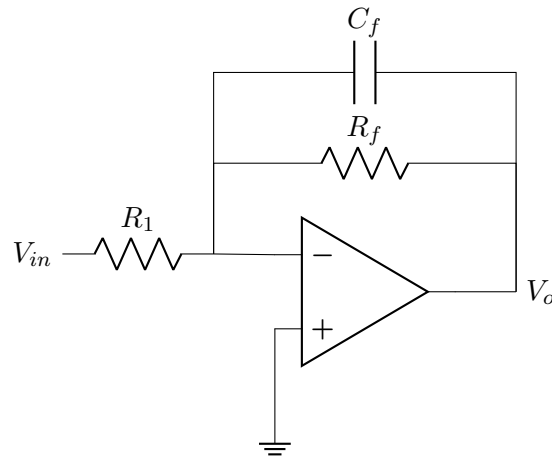


Figure 7: Practical integrator with a finite DC-feedback resistor.

The feedback impedance is

$$Z_f = R_f \parallel \frac{1}{sC_f} = \frac{R_f}{1 + sR_fC_f}.$$

Hence,

$$H(s) = -\frac{R_f/R_1}{1 + sR_fC_f}.$$

Define

$$\omega_c = \frac{1}{R_fC_f}, \quad f_c = \frac{1}{2\pi R_fC_f}.$$

Then

$$H(s) = -\frac{R_f}{R_1} \frac{1}{1 + s/\omega_c}.$$

For $\omega \ll \omega_c$,

$$H \approx -\frac{R_f}{R_1},$$

so the DC gain is finite. For $\omega \gg \omega_c$,

$$H \approx -\frac{R_f/R_1}{j\omega R_fC_f} = -\frac{1}{j\omega R_1C_f},$$

which is the ideal integrator expression.

2(e) Low-pass frequency response

At $s = j\omega$,

$$|H(j\omega)| = \frac{R_f/R_1}{\sqrt{1 + (\omega R_fC_f)^2}} = \frac{R_f/R_1}{\sqrt{1 + (f/f_c)^2}}.$$

Thus:

- for $f \ll f_c$, $|H| \approx R_f/R_1$ (constant pass-band gain);
- at $f = f_c$, the gain is $1/\sqrt{2}$ of its low-frequency value (-3 dB);
- for $f \gg f_c$, the magnitude falls at -20 dB/decade.

Therefore the practical integrator is mathematically a first-order inverting low-pass filter, and it behaves as an integrator above its corner frequency.

Question 3 - First-order all-pass phase shifter

3(a) Derivation using superposition

In the given circuit $R_f = R_1$. Let the voltage at both op-amp inputs be V_x . The non-inverting input receives the output of an RC low-pass network, so

$$V_x = V_+ = \frac{1}{1 + sRC} V_{in}.$$

Because of negative feedback,

$$V_- = V_+ = V_x.$$

At the inverting node, no current enters the op-amp. Therefore,

$$\frac{V_{in} - V_x}{R_1} + \frac{V_o - V_x}{R_f} = 0.$$

Since $R_f = R_1$,

$$V_{in} - V_x + V_o - V_x = 0,$$

so

$$V_o = 2V_x - V_{in}.$$

Substituting $V_x = V_{in}/(1 + sRC)$,

$$\frac{V_o}{V_{in}} = \frac{2}{1 + sRC} - 1 = \frac{1 - sRC}{1 + sRC}.$$

Thus

$$T(s) = \frac{1 - sRC}{1 + sRC}.$$

Let $s = j2\pi f$ and

$$f_c = \frac{1}{2\pi RC}.$$

Then $sRC = jf/f_c$, giving

$$T(jf) = \frac{1 - j(f/f_c)}{1 + j(f/f_c)}.$$

3(b) Amplitude response and all-pass behaviour

The magnitude is

$$|T(jf)| = \frac{|1 - j(f/f_c)|}{|1 + j(f/f_c)|}.$$

Both numerator and denominator have the same magnitude:

$$|1 \pm j(f/f_c)| = \sqrt{1 + (f/f_c)^2}.$$

Therefore,

$$|T(jf)| = 1 \quad \text{for all } f.$$

The circuit passes all frequencies with unity magnitude while changing only the phase. It is therefore a **first-order all-pass filter**.

3(c) Phase response and phase-shifter type

The numerator angle is

$$-\tan^{-1}(f/f_c),$$

and the denominator angle is

$$+\tan^{-1}(f/f_c).$$

Therefore,

$$\phi(f) = -2 \tan^{-1} \left(\frac{f}{f_c} \right).$$

Important values are:

f	ϕ
0	0°
f_c	-90°
∞	-180°

The output phase becomes increasingly negative as frequency increases. Hence the circuit is a **lagging first-order all-pass phase shifter**.

3(d) Frequency for a 90° phase shift

A phase shift of -90° occurs when

$$-2 \tan^{-1}(f/f_c) = -90^\circ.$$

Thus

$$\tan^{-1}(f/f_c) = 45^\circ \Rightarrow f = f_c.$$

Given

$$R = 2 \text{ k}\Omega, \quad C = 0.01 \text{ }\mu\text{F} = 10 \text{ nF},$$

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(2000)(10 \times 10^{-9})}.$$

Therefore,

$$f_c \approx 7.96 \text{ kHz}.$$

The value of R_1 does not set the phase corner; it is required to equal the feedback resistor so that the all-pass transfer function is obtained.

Question 4 - Passive and active first-order filters

4(a) Passive RC low-pass and high-pass filters

Passive low-pass filter

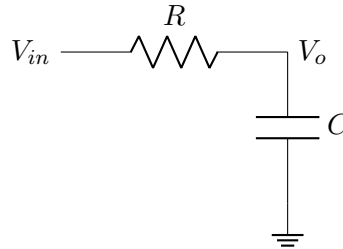


Figure 8: First-order passive RC low-pass filter.

Using the voltage divider,

$$H_{LP}(s) = \frac{Z_C}{R + Z_C}, \quad Z_C = \frac{1}{sC}.$$

Thus

$$H_{LP}(s) = \frac{1}{1 + sRC}.$$

The cut-off angular frequency and frequency are

$$\omega_c = \frac{1}{RC}, \quad f_c = \frac{1}{2\pi RC}.$$

Passive high-pass filter

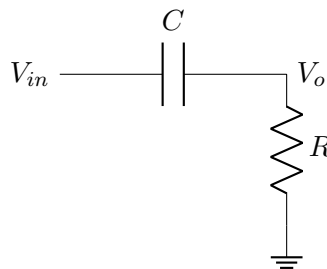


Figure 9: First-order passive RC high-pass filter.

The output is across the resistor:

$$H_{HP}(s) = \frac{R}{R + 1/(sC)}.$$

Therefore,

$$H_{HP}(s) = \frac{sRC}{1 + sRC}.$$

Again,

$$\omega_c = \frac{1}{RC}, \quad f_c = \frac{1}{2\pi RC}.$$

4(b) Active non-inverting low-pass and high-pass filters

Place a non-inverting op-amp after each passive RC network. Because the op-amp input resistance is very high, it does not significantly load the RC network. Let

$$K = 1 + \frac{R_f}{R_g}.$$

Then

$$H_{LP,active}(s) = \frac{K}{1 + sR_L C_L}$$

and

$$H_{HP,active}(s) = K \frac{sR_H C_H}{1 + sR_H C_H}.$$

The cut-off frequencies remain

$$f_L = \frac{1}{2\pi R_H C_H}$$

for the high-pass section and

$$f_H = \frac{1}{2\pi R_L C_L}$$

for the low-pass section. The op-amp adds gain and buffering but does not change the ideal RC corner frequency.

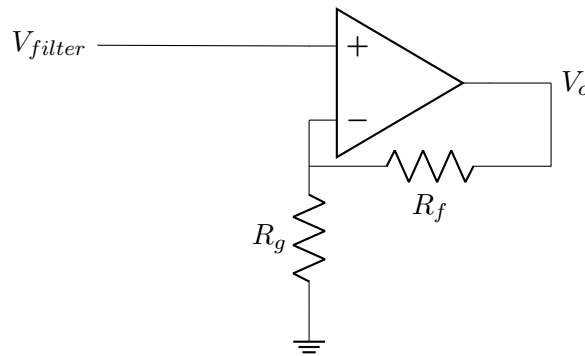


Figure 10: Non-inverting gain stage used with either RC filter.

4(c) Active non-inverting band-pass filter with one gain stage

A wide-band first-order band-pass filter can be formed by placing a passive high-pass section followed by a passive low-pass section, and then using one non-inverting op-amp to provide buffering and gain.

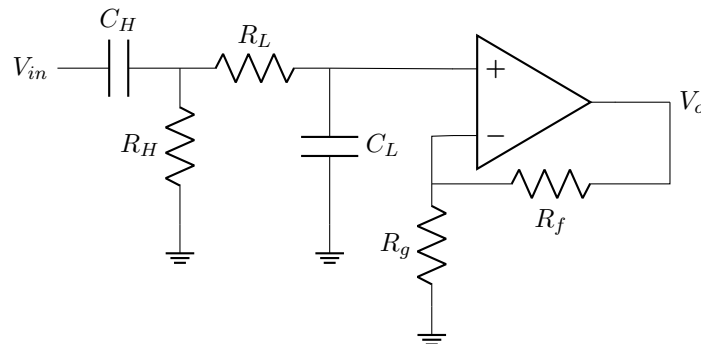


Figure 11: One-op-amp active band-pass filter. The component values should be impedance-scaled to reduce loading between the two passive sections.

Let the non-inverting gain be

$$K = 1 + \frac{R_f}{R_g}.$$

If the two RC sections are treated as effectively isolated, the standard band-pass transfer function is

$$H_{BP}(s) = K \left(\frac{sR_H C_H}{1 + sR_H C_H} \right) \left(\frac{1}{1 + sR_L C_L} \right)$$

or

$$H_{BP}(s) = K \frac{sR_H C_H}{(1 + sR_H C_H)(1 + sR_L C_L)}.$$

The approximate lower and upper cut-off frequencies are

$$f_L = \frac{1}{2\pi R_H C_H}, \quad f_H = \frac{1}{2\pi R_L C_L}, \quad f_L < f_H.$$

For the exact unbuffered passive cascade shown, nodal analysis gives

$$\frac{V_+}{V_{in}} = \frac{sR_H C_H}{1 + s(R_H C_H + R_L C_L + R_H C_L) + s^2 R_H R_L C_H C_L}.$$

The additional term $R_H C_L$ is caused by loading. It becomes small when $R_L \gg R_H$. The total exact response is $V_o/V_{in} = K(V_+/V_{in})$.

At very low frequency, the input capacitor blocks the signal and the response rises at approximately 20 dB/decade. In the pass band, the gain is approximately K . At very high frequency, the shunt capacitor attenuates the signal and the response falls at approximately -20 dB/decade.

4(d) Design for 1 kHz to 10 kHz pass band and gain 2

Required values:

$$f_L = 1 \text{ kHz}, \quad f_H = 10 \text{ kHz}, \quad K = 2.$$

To reduce inter-stage loading, choose $R_L \approx 10R_H$ and select convenient capacitors.

For the high-pass section, choose

$$C_H = 100 \text{ nF}.$$

Then

$$R_H = \frac{1}{2\pi f_L C_H} = \frac{1}{2\pi(1000)(100 \times 10^{-9})} \approx 1.59 \text{ k}\Omega.$$

Use

$$R_H = 1.6 \text{ k}\Omega, \quad C_H = 100 \text{ nF}.$$

The practical corner is approximately

$$f_L \approx \frac{1}{2\pi(1.6 \text{ k}\Omega)(100 \text{ nF})} \approx 995 \text{ Hz}.$$

For the low-pass section, choose

$$C_L = 1 \text{ nF}.$$

Then

$$R_L = \frac{1}{2\pi f_H C_L} = \frac{1}{2\pi(10^4)(10^{-9})} \approx 15.9 \text{ k}\Omega.$$

Use

$$R_L = 15.9 \text{ k}\Omega, \quad C_L = 1 \text{ nF}.$$

This also gives $R_L/R_H \approx 10$, reducing loading.

For gain 2,

$$2 = 1 + \frac{R_f}{R_g} \Rightarrow R_f = R_g.$$

Choose

$$R_f = 10 \text{ k}\Omega, \quad R_g = 10 \text{ k}\Omega.$$

Final design

$$R_H = 1.6 \text{ k}\Omega, \quad C_H = 100 \text{ nF}, \quad R_L = 15.9 \text{ k}\Omega, \quad C_L = 1 \text{ nF}, \quad R_f = R_g = 10 \text{ k}\Omega$$

This gives an approximate pass band of 1 kHz to 10 kHz and a mid-band non-inverting gain of 2.

Question 5 - Comparators and soil-moisture control

5(a) Inverting and non-inverting comparator modes

An analogue comparator is normally operated open-loop. Because the open-loop gain is very high, even a small difference between the two inputs drives the output to one of its saturation levels.

Non-inverting comparator

Apply V_{in} to the non-inverting input and V_{ref} to the inverting input:

$$V_o = \begin{cases} +V_{sat}, & V_{in} > V_{ref}, \\ -V_{sat}, & V_{in} < V_{ref}. \end{cases}$$

The output goes high when the input exceeds the reference.

Inverting comparator

Apply V_{in} to the inverting input and V_{ref} to the non-inverting input:

$$V_o = \begin{cases} +V_{sat}, & V_{in} < V_{ref}, \\ -V_{sat}, & V_{in} > V_{ref}. \end{cases}$$

The output goes low when the input exceeds the reference.

Mode	Input connection	Output-high condition
Non-inverting	$V_{in} \rightarrow +, V_{ref} \rightarrow -$	$V_{in} > V_{ref}$
Inverting	$V_{in} \rightarrow -, V_{ref} \rightarrow +$	$V_{in} < V_{ref}$

5(b) Zero-level and voltage-threshold detectors

Zero-level detector

A zero-level or zero-crossing detector is a comparator with

$$V_{ref} = 0.$$

For a non-inverting zero-crossing detector,

$$V_o = +V_{sat} \quad \text{for } V_{in} > 0,$$

$$V_o = -V_{sat} \quad \text{for } V_{in} < 0.$$

For the inverting form, these output polarities are reversed. The output changes state whenever the input crosses zero.

Voltage-threshold detector

A nonzero reference is applied to one input, usually from a resistor divider or a Zener reference. The comparator changes state when

$$V_{in} = V_{ref}.$$

For example, for a non-inverting threshold detector,

$$V_o = +V_{sat} \quad \text{when } V_{in} > V_{ref},$$

whereas an inverting threshold detector gives the opposite output logic.

In practical circuits, a small amount of positive feedback may be added to create hysteresis. This prevents rapid switching or “chatter” when a noisy input is close to the threshold.

5(c)(i) Circuit to obtain the average soil-moisture voltage

Let the four positive sensor voltages be V_1, V_2, V_3, V_4 . Connect each sensor to a common node through an equal resistor R , and buffer the node with a voltage follower.

Because the op-amp input current is approximately zero, KCL at the averaging node gives

$$\frac{V_1 - V_x}{R} + \frac{V_2 - V_x}{R} + \frac{V_3 - V_x}{R} + \frac{V_4 - V_x}{R} = 0.$$

Therefore,

$$V_1 + V_2 + V_3 + V_4 - 4V_x = 0,$$

so

$$V_x = \frac{V_1 + V_2 + V_3 + V_4}{4}.$$

The voltage follower gives $V_{avg} = V_x$. Thus

$$V_{avg} = \frac{V_1 + V_2 + V_3 + V_4}{4}.$$

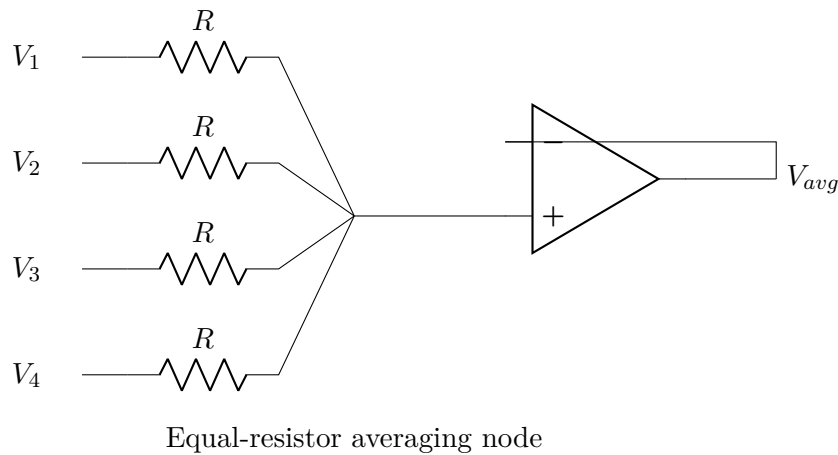


Figure 12: Four-input averaging circuit followed by a voltage follower.

If the sensors have appreciable output resistance, each sensor should first be buffered so that the averaging resistors do not load or cross-couple the sensors.

5(c)(ii) Automatic control of the four water-spraying systems

Use four identical comparator channels. For field i :

- connect V_{avg} to the non-inverting input;
- connect the corresponding sensor voltage V_i to the inverting input.

Then

$$V_{cmp,i} = +V_{sat} \quad \text{when } V_{avg} > V_i.$$

This is exactly the required condition: the field is drier than the average, so its spray system must turn on.

The control law is

$$\text{Spray}_i = \begin{cases} \text{ON}, & V_i < V_{avg}, \\ \text{OFF}, & V_i \geq V_{avg}. \end{cases}$$

Each comparator output should drive a transistor or logic-level MOSFET, which energizes the pump, valve, or relay for that field. If an inductive relay or solenoid is used, a flyback diode must be connected across its coil.

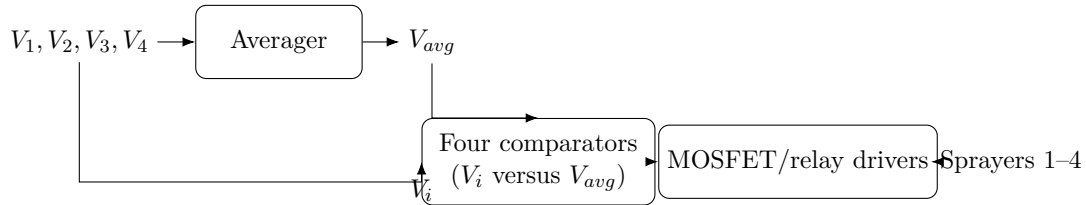


Figure 13: Functional arrangement of the four-field control system.

A small hysteresis band is recommended so that a sprinkler does not rapidly switch on and off when V_i is almost equal to V_{avg} . For example, turn on at $V_i < V_{avg} - \Delta V$ and turn off at $V_i > V_{avg} + \Delta V$.

Question 6 - Wien-bridge network and oscillator

6(a) Transfer function and resonant frequency

The series arm of the Wien network is

$$Z_s = R_1 + \frac{1}{sC_1} = \frac{1 + sR_1C_1}{sC_1}.$$

The shunt arm is

$$Z_p = R_2 \parallel \frac{1}{sC_2} = \frac{R_2}{1 + sR_2C_2}.$$

The network is a voltage divider, hence

$$\beta(s) = \frac{V_o}{V_{in}} = \frac{Z_p}{Z_s + Z_p}.$$

Substitution and simplification give

$$\beta(s) = \frac{sR_2C_1}{1 + s(R_1C_1 + R_2C_1 + R_2C_2) + s^2R_1R_2C_1C_2}.$$

For

$$R_1 = R_2 = R, \quad C_1 = C_2 = C,$$

$$\beta(s) = \frac{sRC}{1 + 3sRC + s^2R^2C^2}.$$

Set $s = j\omega$:

$$\beta(j\omega) = \frac{j\omega RC}{1 - \omega^2 R^2 C^2 + j3\omega RC}.$$

The phase shift is zero when the real part of the denominator is zero:

$$1 - \omega_0^2 R^2 C^2 = 0.$$

Therefore,

$$\omega_0 = \frac{1}{RC}, \quad f_0 = \frac{1}{2\pi RC}.$$

At $\omega = \omega_0$,

$$\beta(j\omega_0) = \frac{j}{j3} = \frac{1}{3}.$$

Thus the Wien network has zero phase shift and an attenuation factor of 1/3 at the resonant frequency.

6(b) Wien-bridge oscillator and conditions for oscillation

The Wien network is connected from the op-amp output to the non-inverting input, forming frequency-selective positive feedback. A resistive divider provides negative feedback and sets the amplifier gain.

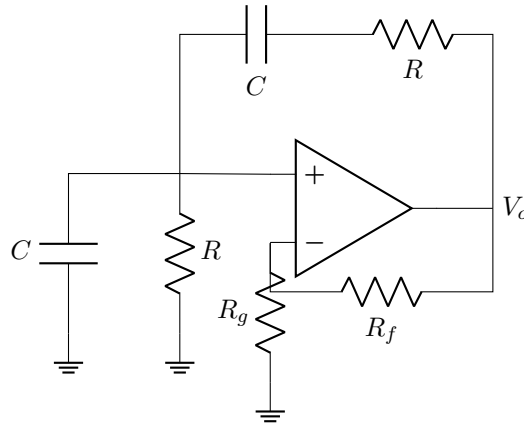


Figure 14: Op-amp Wien-bridge oscillator with frequency-selective positive feedback and resistive negative feedback.

The Barkhausen conditions at the oscillation frequency are:

1. Total loop phase shift must be 0° (or an integer multiple of 360°).
2. Loop-gain magnitude must be unity:

$$|A\beta| = 1.$$

At f_0 , the Wien network gives

$$\beta = \frac{1}{3}.$$

Therefore the non-inverting amplifier must have

$$A = 3.$$

Since

$$A = 1 + \frac{R_f}{R_g},$$

$$1 + \frac{R_f}{R_g} = 3 \quad \Rightarrow \quad \boxed{R_f = 2R_g}.$$

For reliable startup, the small-signal gain is normally adjusted slightly above 3. As the amplitude increases, a lamp, back-to-back diodes, JFET, or another automatic gain-control element reduces the gain toward exactly 3. This stabilizes the sinusoidal amplitude and reduces clipping.

6(c) Design for 1 kHz

Choose convenient equal capacitors:

$$C_1 = C_2 = 10 \text{ nF}.$$

Then

$$R_1 = R_2 = R = \frac{1}{2\pi f_0 C} = \frac{1}{2\pi(1000)(10 \times 10^{-9})} \approx 15.915 \text{ k}\Omega.$$

Choose the practical value

$$\boxed{R_1 = R_2 = 15.9 \text{ k}\Omega}, \quad \boxed{C_1 = C_2 = 10 \text{ nF}}.$$

The resulting frequency is

$$f_0 = \frac{1}{2\pi(15.9 \text{ k}\Omega)(10 \text{ nF})} \approx 1.001 \text{ kHz}.$$

For gain 3, choose

$$R_g = 10 \text{ k}\Omega, \quad R_f = 20 \text{ k}\Omega.$$

Then

$$A = 1 + \frac{20}{10} = 3.$$

A practical oscillator can use a $20\text{ k}\Omega$ resistor plus a small trimmer to set startup gain slightly above 3, together with an amplitude-stabilizing element.

A realistic 1 kHz design

$$R_1 = R_2 = 15.9\text{ k}\Omega, \quad C_1 = C_2 = 10\text{ nF}, \quad R_g = 10\text{ k}\Omega, \quad R_f = 20\text{ k}\Omega$$

Use 1% resistors and reasonably matched capacitors for accurate frequency and low distortion.

Question 7 - Op-amp voltage regulators

7(a) Line regulation and load regulation

Line regulation

Line regulation describes how well a regulator maintains a constant output voltage when the input supply voltage changes while the load is held constant. It can be expressed as

$$\text{Line regulation} = \frac{\Delta V_o}{\Delta V_{in}}$$

with units such as mV/V, or as a percentage.

Load regulation

Load regulation describes how well the regulator maintains a constant output voltage when the load current changes while the input voltage is held constant. It is often specified as

$$\Delta V_o = V_{o,NL} - V_{o,FL}$$

or

$$\text{Load regulation}(\%) = \frac{V_{o,NL} - V_{o,FL}}{V_{o,FL}} \times 100\%.$$

Here NL and FL denote no-load and full-load conditions.

7(b) Output voltage of the regulator in Figure 5

The Zener diode establishes approximately

$$V_+ = V_Z = 5.1 \text{ V}.$$

The inverting input senses the divided output voltage:

$$V_- = V_o \frac{R_3}{R_2 + R_3}.$$

Negative feedback forces

$$V_- = V_+ = 5.1 \text{ V}.$$

Therefore,

$$5.1 = V_o \frac{R_3}{R_2 + R_3}.$$

Hence,

$$V_o = 5.1 \left(\frac{R_2 + R_3}{R_3} \right) = 5.1 \left(1 + \frac{R_2}{R_3} \right).$$

Given

$$R_2 = R_3 = 10 \text{ k}\Omega,$$

$$V_o = 5.1(1 + 1) = 10.2 \text{ V}.$$

Thus

$$V_{OUT} = 10.2 \text{ V}.$$

The pass transistor's base-emitter voltage does not need to be subtracted from this result because the feedback is taken from the actual output after the transistor. The op-amp automatically supplies the necessary extra base voltage, provided it has sufficient output swing and the input supply has adequate headroom.

7(c) Current limiting at 700 mA

Insert a small current-sense resistor R_S between the emitter of the series pass transistor and the regulated output. Add a small NPN transistor Q_2 so that:

- its base is connected to the pass-transistor side of R_S ;
- its emitter is connected to the load side of R_S ;
- its collector is connected to the drive node of the pass transistor.

When the load current is small, the voltage across R_S is below the base-emitter turn-on voltage and Q_2 is off. As the current approaches the limit,

$$V_{RS} = I_L R_S \approx V_{BE(Q_2)}.$$

Then Q_2 turns on and diverts/clamps the drive to Q_1 , preventing a further increase in load current.

Using $V_{BE} \approx 0.7\text{ V}$,

$$R_S = \frac{V_{BE}}{I_{lim}} = \frac{0.7}{0.7} = 1.0\ \Omega.$$

Therefore choose

$$R_S \approx 1\ \Omega.$$

Its power at the limit is

$$P_{RS} = I_{lim}^2 R_S = (0.7)^2(1) = 0.49\text{ W}.$$

Use at least a 1 W resistor for reliable operation.

If $V_{BE} = 0.6\text{ V}$ is assumed, the calculated value is $0.6/0.7 \approx 0.86\ \Omega$; a nearby standard value can be selected according to the desired tolerance and transistor temperature.

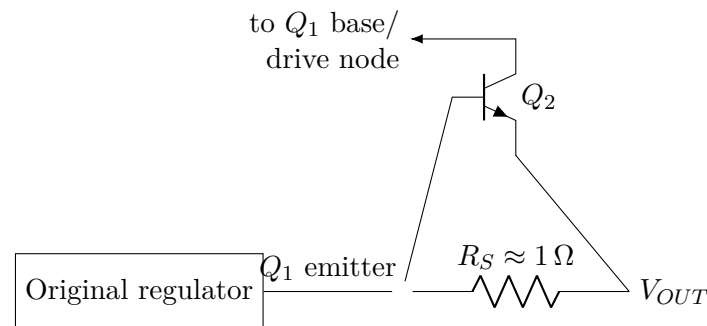


Figure 15: Principle of the current-limiting modification.

7(d) Basic op-amp shunt regulator

A shunt regulator uses a series resistor between the unregulated source and the output, and a controllable transistor connected in parallel with the load.

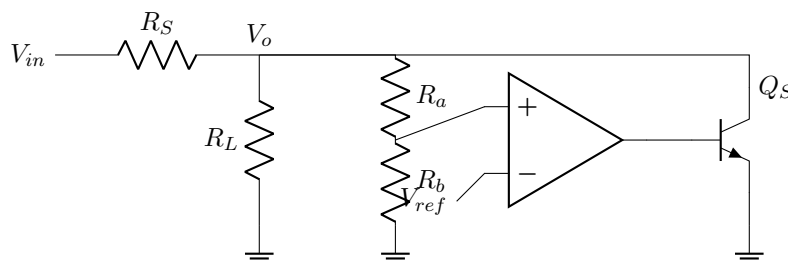


Figure 16: Basic op-amp shunt regulator. The controlled transistor diverts excess current to ground.

The divider senses a fraction of the output:

$$V_s = V_o \frac{R_b}{R_a + R_b}.$$

The op-amp compares this sensed voltage with a stable reference. If V_o rises, the control circuit increases the shunt transistor current. The increased current causes a larger voltage drop across the series resistor R_S , which reduces V_o . If V_o falls, shunt current is reduced, the drop across R_S becomes smaller, and V_o rises. This is negative feedback.

At equilibrium,

$$V_s = V_{ref},$$

so

$$V_{ref} = V_o \frac{R_b}{R_a + R_b}.$$

Therefore,

$$V_o = V_{ref} \left(1 + \frac{R_a}{R_b} \right).$$

The input current through the series resistor is divided between the load and the shunt transistor:

$$I_S = I_L + I_{sh}.$$

The regulator controls I_{sh} so that the output voltage remains approximately constant.

7(e) Inherent short-circuit protection of a shunt regulator

If the output is accidentally short-circuited,

$$V_o \approx 0.$$

The shunt transistor is driven toward cutoff because no extra shunt current is required. The short-circuit current is then limited mainly by the series resistor:

$$I_{SC} \approx \frac{V_{in}}{R_S}$$

(or more precisely $(V_{in} - V_o)/R_S$). Therefore the circuit does not contain a series pass transistor that must carry an uncontrolled short-circuit current. The series resistor inherently limits the maximum current.

The series resistor must nevertheless be rated for the worst-case short-circuit power:

$$P_{R_S, SC} \approx \frac{V_{in}^2}{R_S}.$$

This simple current limiting is the reason a basic shunt regulator has inherent short-circuit protection, although it is less power-efficient than a series regulator during normal operation.

Final revision summary

Topic	Essential result
Negative feedback	$A_f = A/(1 + A\beta) \approx 1/\beta$ when $A\beta \gg 1$
Inverting amplifier	$V_o = -(R_f/R_1)V_{in}$
Non-inverting amplifier	$V_o = (1 + R_f/R_1)V_{in}$
True integrator	$H(s) = -1/(sR_1C_f)$
Practical integrator	$H(s) = -(R_f/R_1)/(1 + sR_fC_f)$
All-pass phase shifter	$T(s) = (1 - sRC)/(1 + sRC)$, $ T = 1$, $\phi = -2 \tan^{-1}(f/f_c)$
Passive low-pass	$H = 1/(1 + sRC)$
Passive high-pass	$H = sRC/(1 + sRC)$
Comparator	Output polarity is determined by the sign of $V_+ - V_-$
Four-sensor average	$V_{avg} = (V_1 + V_2 + V_3 + V_4)/4$
Wien network	$f_0 = 1/(2\pi RC)$, $\beta(f_0) = 1/3$
Wien oscillator	Non-inverting gain $A = 3$, hence $R_f = 2R_g$
Figure 5 regulator	$V_o = 5.1(1 + 10k/10k) = 10.2 \text{ V}$
700 mA current limit	$R_S \approx 0.7/0.7 = 1 \Omega$, use at least 1 W